Development of a Novel miRNA-Based Diagnostic Tool Utilizing Toehold Switch Technology for Early Detection of Breast Cancer in Point-of-Care Settings

Ishaan Mehrotra[1], Shubhika Munot[1], Christian Pile[1], Andy Sun[1], Emma J. Chory[1,†], Cameron M. Kim[1,†]

Affiliations
[1]Department of Biomedical Engineering, Duke University, Durham, NC 27705, USA.
[†]Corresponding authors: emma.chory@duke.edu, cameron.kim@duke.edu

ABSTRACT
Breast cancer, accounting for 13% of all cancer deaths in females, remains a leading cause of cancer-related mortality. While early detection greatly improves patient survival rates and long-term outcomes, traditional diagnostic methods such as mammography and biopsy are often limited by cost, accessibility, and patient discomfort. To address this challenge, we have proposed a novel, toehold switch-based tool for detecting the upregulation of breast cancer-associated miRNA biomarkers, specifically miR-155 and miR-10b via a cell-free fluorescence-based reporter system. Toehold switches are synthetic RNA sequences that remain untranslated unless bound by their target, thus making it a compelling biosensor for the precise detection of these nucleic acids. Our proposed point-of-care diagnostic solution, integrates toehold switches with an AND-gated reporter module to ensure high specificity and sensitivity in detecting miRNAs. In silico validation, including secondary structure modeling, thermodynamic interactions, and stability of the computationally designed toehold switches yielded three ideal candidates for experimental testing. Double-transformation assays of the candidate toehold switches with their respective miRNAs confirmed functional activation of toehold switches in response to the target via OD600 normalized quantitative measurements of GFP expression. miR-10b toehold switch 2 and miR-155 toehold switch 7 exhibited the highest activations upon trigger binding. Cloning of the reporter module is still being explored as initial sequencing results yielded unsuccessful integration of our tandem promoters into our pET28a plasmid backbone. Overall, the work aims to develop a highly specific and non-invasive diagnostic system that can detect breast cancer-associated miRNAs with high sensitivity, minimal false positives, and ultimately has the potential to improve early detection in resource-limited settings. Future studies will focus on optimizing assay sensitivity and expanding the detection platform to additional breast cancer-associated miRNAs, enabling a multi-marker diagnostic approach.

GRAPHICAL ABSTRACT

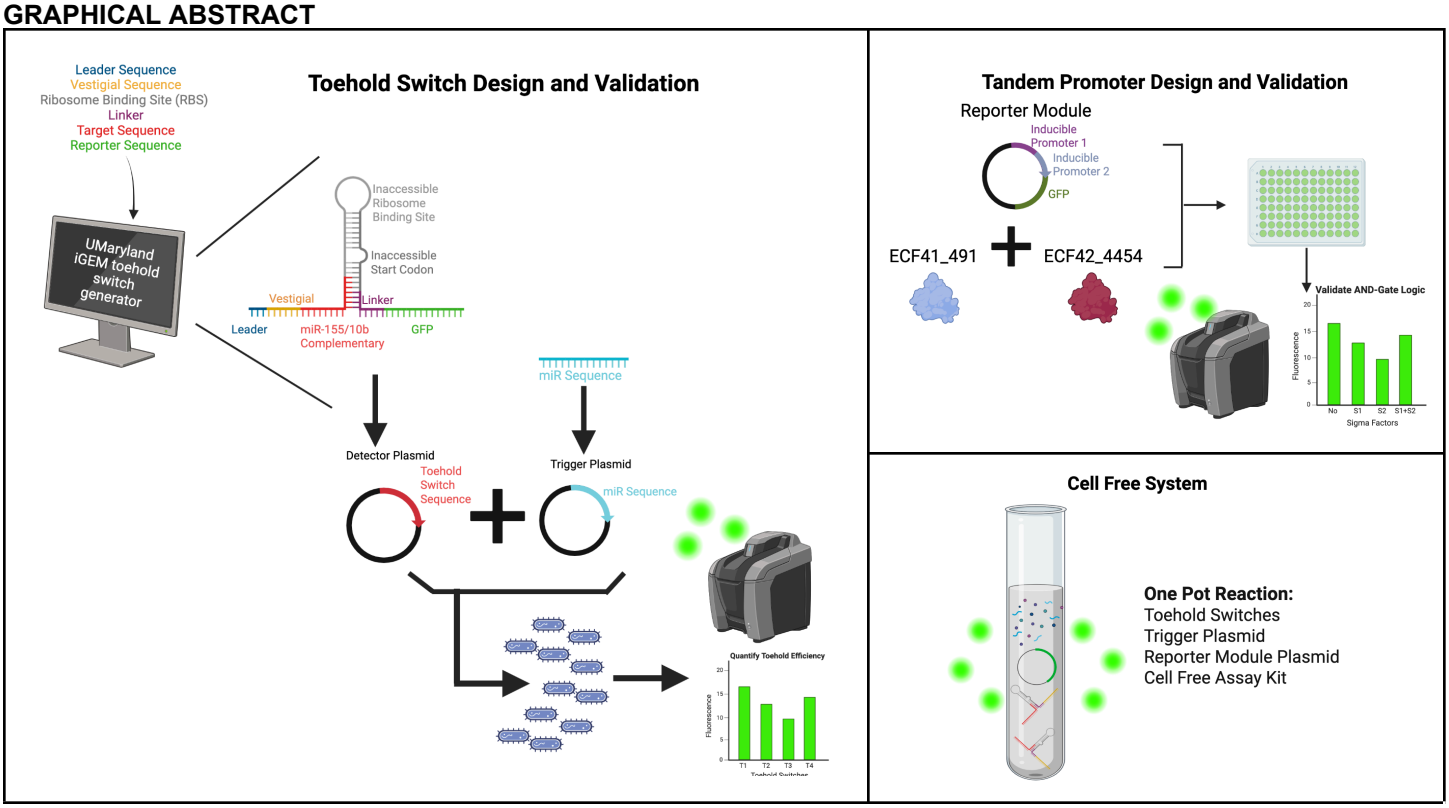


Figure 1: Graphical Abstract of the Major Components of the Diagnostic System

INTRODUCTION

Breast cancer claims nearly 700,000 lives annually worldwide, yet early diagnosis remains one of the most effective ways to improve survival and outcomes. In fact, breast cancer is the second leading cause of death as it relates to cancer in women across the US [1]. On a global scale, 2.3 million cases of breast cancer were diagnosed in 2020 and it is estimated that by 2040, 3 million women will be diagnosed with breast cancer and the annual death toll will reach 1 million [2]. However, since 1989, breast cancer death rates have shown a steady decrease of 42% in the US due to earlier screening, increased awareness and improved treatment technologies [1].

As a result, early diagnosis of breast cancer plays a critical role in improving patient outcomes. According to the American Cancer Society, a patient's 5-year relative survival rate is 99% percent when breast cancer is detected early and caught in the localized stages (i.e. stage 0, I, and II). However in late stage diagnoses, a patient's 5-year relative survival rate is between approximately 30% and 85%, with survival rate decreasing as detection stage increases [3]. Moreover, there is a significant financial burden on patients as the detection stage increases. On average, patients diagnosed at early stages (0, I or II) have treatment costs of approximately $60,000-$83,000, but patients diagnosed at later stages (III, and IV) have approximate treatment costs of $130,000-$135,000 [4]. Overall, these statistics underscore the importance of regular screening and early detection.

Early diagnosis has caught stage I-II cancer in 70% of breast cancer patients living in high-income countries, but in low to middle income countries a majority of breast cancer patients remain undiagnosed until late stages, thus contributing to poor patient prognosis [5]. Diagnostic disparities are not limited to country income, but also arise from access to healthcare due to regional landscapes. For example, countries along the Appalachian mountains in the United States have significantly higher mortality rates for breast cancer [6]. While modifiable secondary factors such as obesity, birthing patterns, and substance use are accountable for some of this higher mortality, major causes for an increase in breast cancer related mortality are due to inaccessibility of screening and late diagnoses [6].

Current breast cancer diagnostic strategies primarily rely on imaging technologies such as mammography, ultrasound, and MRI, followed by tissue biopsy for confirmation. While mammography remains the gold standard for screening, this technology experiences notable limitations including false positives, radiation exposure, and reduced sensitivity in dense breast tissue. Additionally, traditional diagnostic processes often require specialized facilities and trained personnel, creating accessibility barriers particularly in rural and low-income areas. Although there have been innovations in these imaging techniques to facilitate earlier detection and accuracy, there has been significant interest in the area of liquid biopsies which involves analyzing blood or other bodily fluids for tumor biomarkers. This is due to the fact that many imaging diagnostic tools are only available in areas with well developed healthcare infrastructure whereas liquid biopsies have the potential to expand access to screening by contributing to the development of point of care devices in resource limited settings. To this end, our work focuses on increasing the accessibility of breast cancer screening for underserved populations.

In various cancers, microRNAs (miRNA) are highly dysregulated via amplification of miRNA genes, abnormal transcription, and epigenetic changes in miRNA machinery [7]. Genes involved in proliferative signaling, cell death, growth suppression, metastasis, and angiogenesis are all impacted by miRNA. As such, miRNA can function as tumor suppressors as well as oncogenes [8]. The clinical application of microRNA-based diagnostics has gained momentum in recent years, with several detection methods established across healthcare settings. Currently, quantitative reverse transcription PCR (qRT-PCR) represents the gold standard for microRNA detection, offering high sensitivity but requiring specialized laboratory equipment [9]]. Next-generation sequencing provides comprehensive profiling but remains costly and time-intensive [10]. More accessible alternatives include microarray platforms and the emerging electrochemical biosensors, which offer point-of-care potential at reduced costs. Despite technical advances, only a limited number of microRNA diagnostics have received FDA approval, with miRView Lung (for lung cancer classification) and miRView Mets (for identifying tumor origin) being notable examples [11]. The accessibility of these tests remains restricted primarily to specialized oncology centers, highlighting the need for more widely deployable diagnostic tools.

This has led to significant research into the use of miRNAs as biomarkers for cancer diagnosis. In our research, we have identified two major miRNA biomarkers indicative of breast cancer: miR-155 and miR-10b. Firstly, increased miRNA-155 expression has been shown to promote tumor cell proliferation and inhibit apoptosis by targeting PTEN (tumor suppression), PI3K (cell proliferation), and AKT (cell proliferation) [8]. Moreover, previous research has also identified miR-155 as a suitable biomarker for breast cancer due to high specificity, AUC, and sensitivity [12]. Secondly, overexpression of miR-10b is primarily associated with the TWIST-1 pathway which leads to high CD44 and LMP1 expression. CD44 is a protein involved in cell adhesion and migration, while LMP1 can trigger apoptosis prevention pathways and downregulate E-Cadherins (a critical component of cell adhesion) [13]. This combination leads to a suitable environment for metastasis, and as such miR-10b has been identified as a biomarker, specifically for metastatic breast cancer [13].

With these specific biomarkers in mind, toehold switches emerged as a suitable diagnostic tool. Toehold switches have been used in a variety of diagnostic applications including but not limited to Ebola virus, E.coli, norovirus, and Zika detection [19]. Previous work has also showcased the effectiveness of using miRNA sensors for the detection of miRNA in mammalian cells [14].

Therefore, this work aims to develop an innovative toehold switch based diagnostic tool for the early detection of breast cancer, leveraging the miR-155 and miR-10b as biomarkers. At its core, toehold switches function as programmable RNA sensors that regulate translation through a conformational change mechanism. In their default state, these synthetic RNA structures form a hairpin that sequesters the ribosome binding site (RBS) and start codon within the stem region, effectively preventing translation of a downstream reporter gene. Each switch contains a toehold domain—a single-stranded region at the 5' end that serves as the initial binding site for the target miRNA. When the specific target miRNA binds to this toehold domain, it initiates strand displacement through branch migration, unwinding the hairpin structure and exposing the previously sequestered RBS and start codon. This conformational change enables ribosome access and subsequent translation of the reporter gene, producing a detectable output signal.

The primary objectives of this research are to design and synthesize three distinct toehold switches for both miR-155 and miR-10b, and to test their efficacy in an E. coli model system. This will involve evaluating the switches' ability to accurately detect the presence of these miRNAs under laboratory conditions. Additionally, the project seeks to validate the AND logic gate system by designing a reporter module that utilizes a tandem promoter to ensure that the expression of the reporter gene—GFP—is contingent upon the simultaneous detection of both miRNA targets. This work validated that the toehold switch-based system will provide a highly sensitive method for detecting miR-155 and miR-10b in a non-invasive manner. This has the potential to significantly improve early breast cancer detection, especially in low-resource settings, thus enhancing the survival rates and quality of life for patients. The successful development and optimization of this diagnostic tool could lead to its application in routine clinical practice, offering a cost-effective, accessible, and patient-friendly alternative to current diagnostic methods.

RESULTS

Computational Validation and Folding Simulations of Toehold Switches

To identify the most suitable toehold switch candidates with high specificity for miR-10b and miR-155 among our library of designed toehold switches, computational validation was performed using three algorithms: uMaryland iGEM, RNAcofold, and RNAup. Results from these algorithms are summarized in **Table 1 and Table 2** for miR-10b and miR-155 respectively. The tables present the change in minimum free energy, ΔG of binding, and total free energy of binding values for target interactions. ΔG of binding for off-target interactions are also shown in the tables to account for off-target interactions of the toehold switches. This analysis allowed optimal candidates to be selected on the basis of comparatively low ΔG values for the target miRNA interactions and a comparatively high ΔG value for off-target interactions. These two metrics facilitated a quantitative measurement of specificity and strong binding affinity. miR-10b toeholds **2, 5, and 7** demonstrated optimal characteristics (Table 1), while miR-155 toeholds **4, 5, and 7** exhibited similarly favorable properties based on the two aforementioned metrics.

Once these optimal toehold candidates were selected, computational folding simulations were conducted in Dinamelt to verify the structural integrity and correct folding of the toeholds. The results of computational folding are displayed in **Figure 1** and these corroborated the initial findings of the computational validation. Specifically, miR-10b toeholds **2**, **5**, and **7 (Figure 1a)**, and miR-155 toeholds **4**, **5**, and **7 (Figure 1b)** consistently showed favorable folding predictions, further substantiating the selection of these switches.

Plasmid Design for De Novo Plasmid with Toeholds

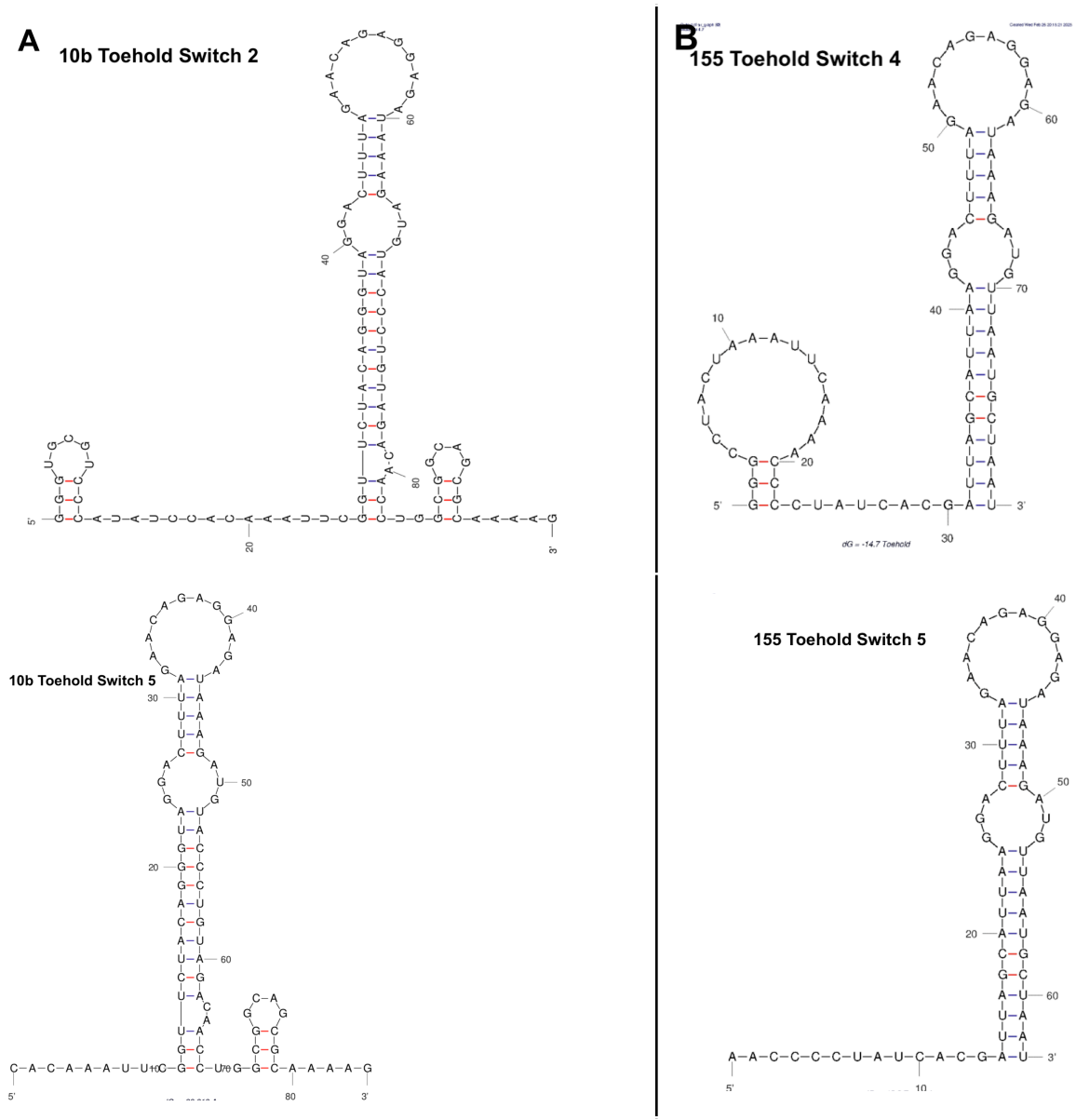
To enable functional validation of the selected toehold switches in vitro, de novo plasmids were designed and assembled. **Figure 2** presents the annotated schematic highlighting critical components such as the promoter, RBS, and the toehold switch module itself, to ensure specific activation by the miR. **Figure 2(a)** provides a linear layout of a representative construct (miR-155 T5), while **Figure 2(b)** illustrates the annotated high level plasmid.

miR-10b Toehold Switch Number	Leader Sequence	Vestigial	iGEM	RNAcofold	RNAup	Off-Target Effects	
			ΔMFE (kcal/mol)	ΔG of binding (kcal/mol)	Total FE of binding (kcal/mol)	miR-21 - ΔG of binding (kcal/mol)	miR-155 - ΔG of binding (kcal/mol)
1	No	GUUACUGUAUGACC	-23.51	-21.98	-24.1	-	-
2	No	GGGUGCGUCCCAUAUC	-25.25	-22.55	-26.24	-4.82	-6.84
3	No	GGGAUACUAUUUAACGAA	-22.64	-22.21	-24.89	-	-
4	No	GGGCCUACUAAAUUCAA	-22.13	-21.81	-23.77	-	-
5	No	None	-25.09	-25.07	-24.96	-7.48	-9.10
6	Yes	None	-23.27	-22.53	-22.47	-	-
7	Yes	None(Liu template)		-21.2	-21.65	-7.48	-9.10

Table 1: Computational Validation of de novo designed miR-10b toehold switches using three separate algorithms (uMaryland iGEM software, RNAcofold, RNAup) and “off target” interaction analysis with miR-21 and miR-155.

miR-155 Toehold Switch Number	Leader Sequence	Vestigial	iGEM	RNAcofold	RNAup	Off-Target Effects	
			ΔMFE (kcal/mol)	ΔG of binding (kcal/mol)	Total FE of binding (kcal/mol)	miR-21 - ΔG of binding (kcal/mol)	miR-10b - ΔG of binding (kcal/mol)
1	No	GUUACUGUAUGACC	-25.86	-23.87	-25.56	-	-
2	No	GGGUGCGUCCCAUAUC	-25.06	-25.22	-26.48	-	-
3	No	GGGAUACUAUUUAACGAA	-24.36	-25.08	-25.1	-	-
4	No	GGGCCUACUAAAUUCAA	-25.3	-25.26	-25.92	-4.66	-7.09
5	No	None	-25.67	-27.58	-27.01	-5.33	-6.6
6	Yes	None	-23.34	-25.55	-25.15	-	-
7	No	None(Liu template)		-29.86	-29.21	-5.28	-7.29

Table 2: Computational Validation of de novo designed miR-155 toehold switches using three separate algorithms (uMaryland iGEM software, RNAcofold, RNAup) and “off target” interaction analysis with miR-21 and miR-10b.



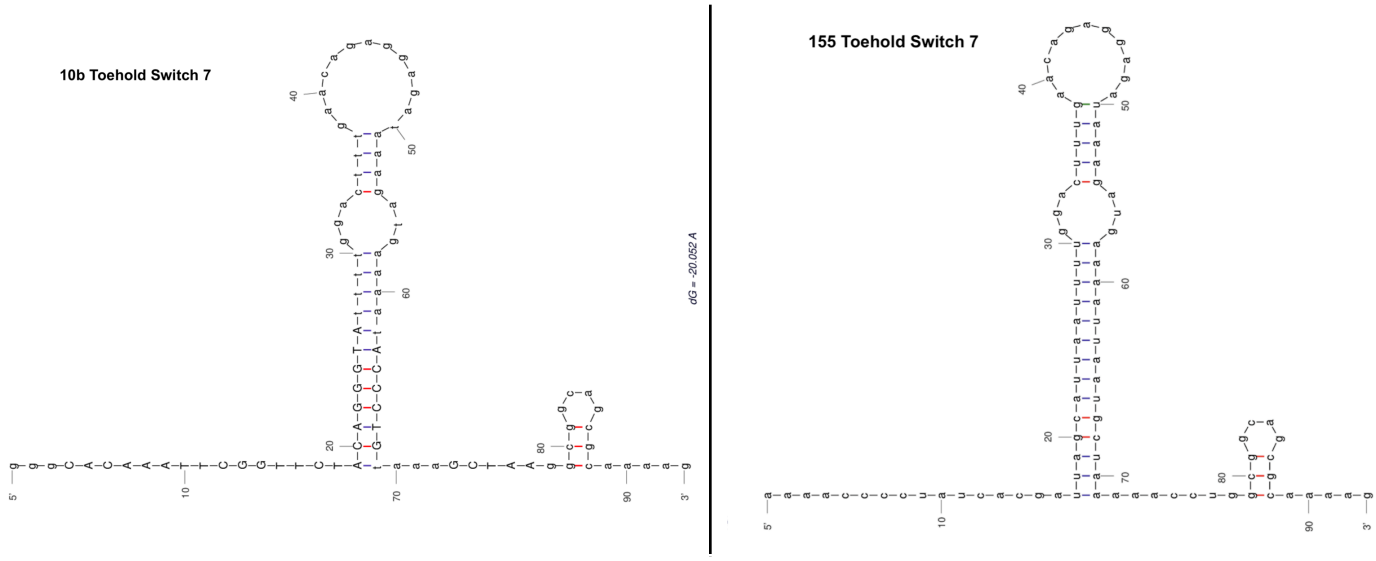


Figure 1: Folding Simulations for selected Toehold switches based on computational validation
Figure 1(A) shows the Dinamelt folding simulations of miR10b toehold switches 2, 5 and 7. Figure 1(B) shows the Dinamelt folding simulations of miR155 toehold switches 4,5,7. These specific toeholds were selected as good candidates based on the computational metrics of ΔMFE,Δ G of binding and Total FE of binding.

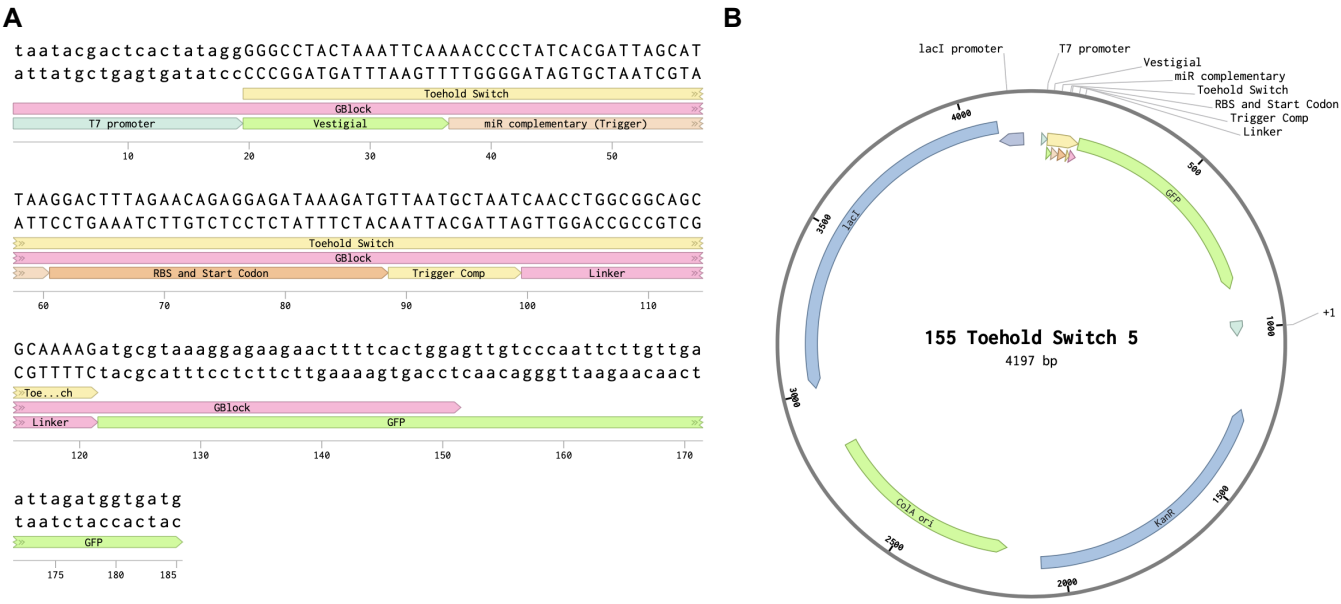


Figure 2: Schematic of Plasmid Design for De Novo Plasmid with Toehold Switches
(A) Linear visualization of the Toehold Switch 5 for miR 155 detection emphasizing the different components of the designed toehold **(B)** Annotated high level plasmid construct. **Supplementary Figure 5** provides a fully annotated plasmid map.

Promoter and RBS verification of De Novo Plasmid Constructs

To ensure the correct assembly and transcriptional efficiency of our designed reporter module plasmid construct, computational validation of the tandem promoter design was conducted via De Novo DNA. This also facilitated the selection of an optimal linker for our tandem promoters in the reporter module design. **Figure 3** illustrates that the linker sequence for our designed construct exhibits robust transcriptional activity and correct functional alignment. Essentially, **Figure 3(a)** highlights the desired unidirectional transcription for the construct while **Figure 3(b)** shows desired minimal reverse transcription. Overall, this demonstrates transcriptional directionality and strength enabled by the RBS and short AT-rich linker in the plasmid construct. These results confirm the precise assembly and functional efficacy of these crucial elements, ensuring the plasmid's ability to correctly function as an 'AND-logic gate' reporter system.

Design for De Novo Plasmid with Tandem Promoters
To achieve precise AND gate logic control of gene expression, plasmids were designed to incorporate tandem promoters (**Figure 4**), to ensure specific and synergistic activation by the corresponding sigma factors. **Figure 4(a)** details the linear arrangement emphasizing the tandem promoters linked via an AT-rich sequence, and **Figure 4(b)** illustrates the annotated high level plasmid construct.

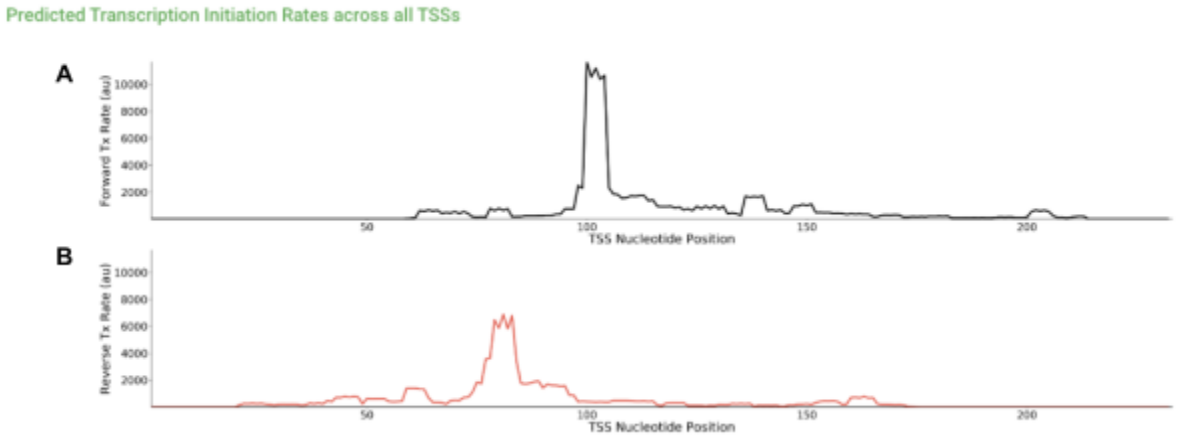


Figure 3: Predicted transcription initiation rates for tandem promoter construct with short AT-rich linker. Transcription start site (TSS) profiling was performed in silico to evaluate the directionality and strength of initiation events across the DNA sequence. The forward strand (top, black) shows a strong, localized initiation peak around nucleotide position ~100, indicating efficient unidirectional transcription. In contrast, the reverse strand (bottom, red) exhibits a weaker and broader peak in the same region, suggesting minimal reverse transcription. These results support the use of a short AT-rich linker, ATTATATAAT to promote focused, unidirectional transcription in tandem promoter systems.

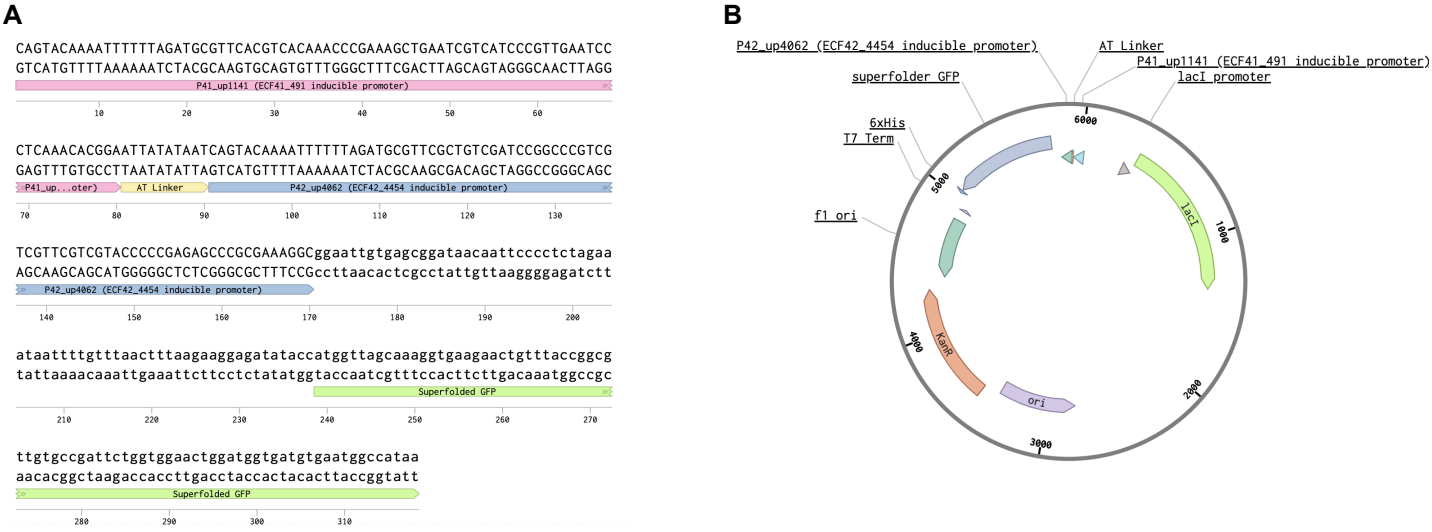
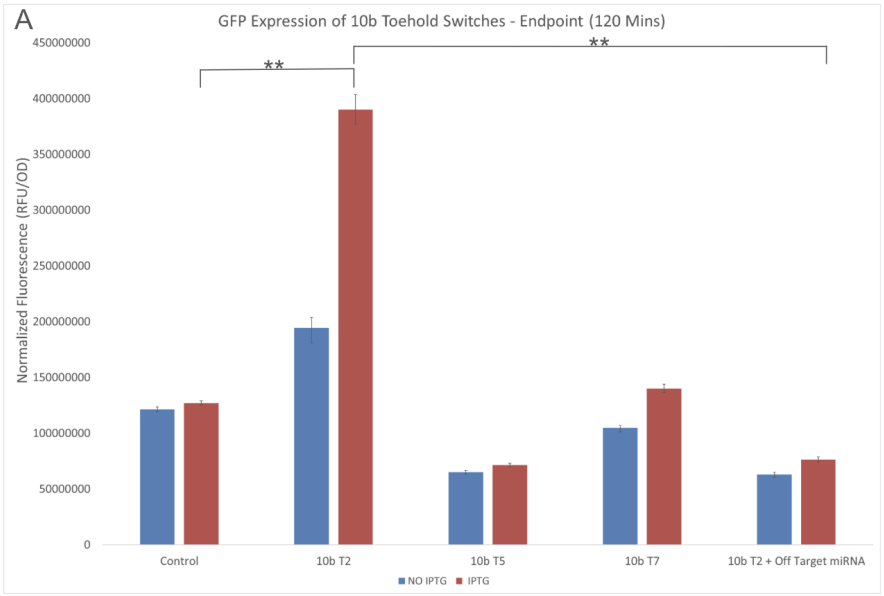


Figure 4: Schematic of Plasmid Design for De Novo Plasmid with Tandem Promoters
(A) Linear visualization of the reporter module emphasizing the integration of tandem promoters configured for AND gate logic, essential for ensuring precise and specific activation of the reporter module in response to targeted miRNA detection. **(B)** Annotated high level plasmid construct. **Supplementary Figure 6** provides a fully annotated plasmid map.

Double Transformation of microRNA plasmids and Toehold switch plasmids in E Coli.
To verify the efficacy and specificity of the 3 selected toehold switch candidates for miR-10b and miR-155, the GFP fluorescence was measured from bacterial colonies which were sequentially double transformed with one of the target miR and their respective toehold switches under IPTG induction. The GFP fluorescence was measured after a time period of 120 minutes and subsequently normalized to OD600 in order to account for differences in cell density among samples. As showcased in **Figure 6(a)**, **miR-10b Toehold switch 2 exhibited the highest level**

of fluorescence in comparison to the control case in which no toehold switch was transformed with the target miR. Conversely, the other candidate toehold switches did not not show significantly higher levels of GFP expression in comparison to the control. Moreover, upon a sequential transformation of an off-target miR with 10b toehold switch 2, the GFP expression was significantly lower in comparison to the control. **Figure 6(b)** showcases that all three miR-155 candidate toehold switches exhibit higher levels of GFP expression in comparison to the control case in which no toehold switch was transformed with miR-155. However, **miR 155 toehold switch 7** exhibited the highest level of fluorescence among all of the toehold switches. Similarly, the off-target experiment yielded results which validated the specificity of 155 toehold switch 7 as the GFP fluorescence was comparable to the fluorescence observed in the control case.



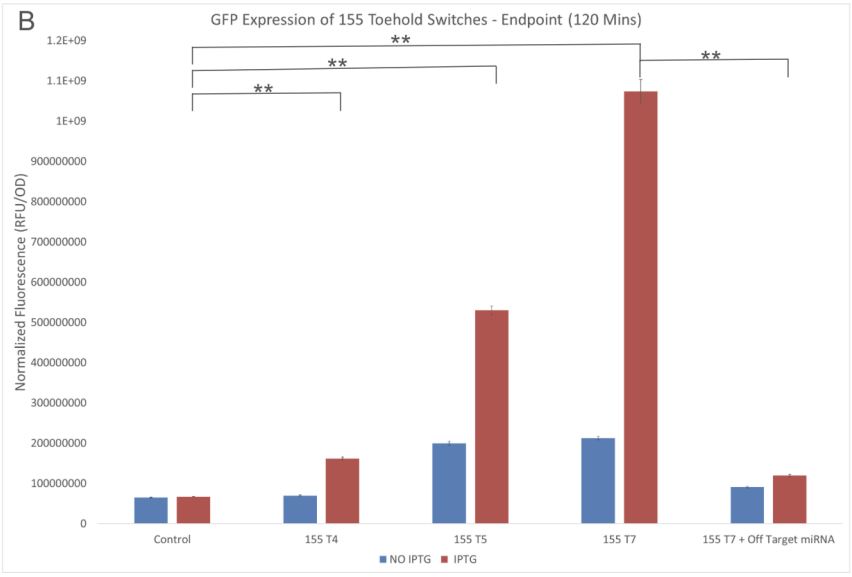


Figure 6: (A) GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and miR-10b plasmids, normalized to OD600. **(B)** The level of GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and miR-155 plasmids, normalized to OD600. Statistical significance is denoted by ** above.

Reporter Module Cloning

To facilitate the generation of an accurate reporter system, we sought to clone an “AND-gate” logic plasmid by incorporating tandem promoters which would be activated to express GFP in the presence of sigma factors ECF41_491 and ECF42_4454. To construct this reporter module, we utilized the pET28a plasmid as a backbone due to its compatibility with T7-based expression systems and its modularity for synthetic circuit development. However, we experienced challenges during the cloning process and were unable to successfully clone our reporter module as seen in the sequence alignment displayed in **Figure 7**.

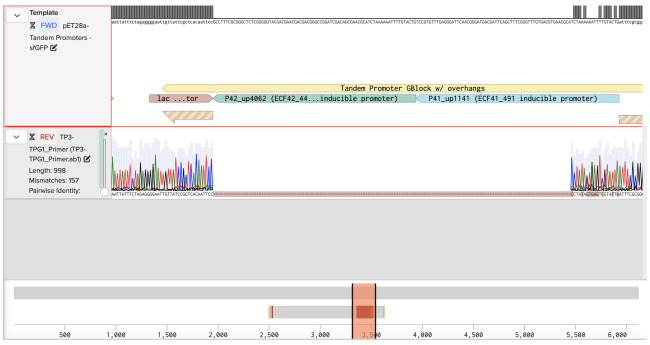


Figure 7: Sequence Alignment Results for Cloning of Tandem Promoter Reporter Module.
Sequence alignment of the constructed reporter plasmid indicated unsuccessful cloning of the tandem promoter-GFP module.

Validation of Cell Free System

To confirm the reliability of our cell-free validation system, a control experiment was conducted using the known EforRed toehold and its corresponding trigger RNA. The testing group showed visible color under RFP readings **(Figure 8**), validating the functionality and sensitivity of our cell-free system. This control experiment provided further confidence in the accuracy and efficacy of subsequent toehold switch evaluations using the cell-free assay.

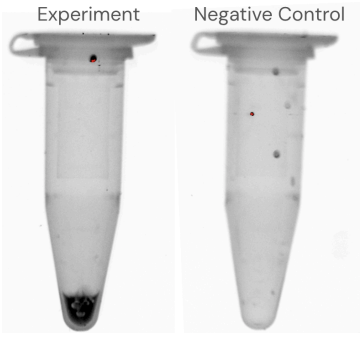


Figure 8: Validation of Cell-Free System Using Known EforRed Toehold and Trigger.
The cell-free transcription-translation system was validated using a known EforRed toehold switch and its complementary trigger miRNA.

DISCUSSION

Computational Validation of the Toehold Switches

The computational analysis and folding simulations performed on the de novo designed miR toehold switches targeting miR-10b and miR-155 provided critical insights into their predicted structural integrity, thermodynamic stability, and binding specificity. The multi-faceted in silico validation approach which combined DinaMelt, RNAcoFold, RNAup, and the UMaryland iGEM toehold generator, allowed for an in-depth assessment of each candidate’s suitability for further experimental testing. The favorable ΔG values observed for several toeholds, particularly miR-10b toeholds 2, 5, and 7 and miR-155 toeholds 4, 5, and 7, indicated strong hybridization between the toehold and the respective target miR. These switches also consistently displayed stable secondary structures in DinaMelt simulations and minimal predicted off-target interactions with unrelated miRNAs such as miR-21. The clear thermodynamic distinction between the free energy of binding to intended versus non-target sequences suggested a high degree of target specificity which is an essential requirement. Additionally, the predicted structures remained stable at physiological temperatures (25°C), which reinforced the potential functional reliability of our designed switches within cell-free systems such as PURExpress. Most notably, the inclusion of leader and different vestigial sequences in our toehold switch designs facilitated the exploration of structural configurations that preserved the hairpin structure while enhancing binding kinetics.

Design and Functional Verification of Plasmid Constructs

By repurposing well-characterized plasmid backbones from Addgene, we leveraged prior validated frameworks, introducing precise modifications in order to facilitate the design of plasmids which were able to express the synthetic toehold switches and plasmids which were able to express their corresponding miR triggers as well. Successful sequencing and alignment analyses confirmed the accurate incorporation of the designed sequences into the plasmid constructs. These sequencing results provided critical evidence of the functional integrity of our engineered constructs, validating our plasmid design strategy and establishing a reliable foundation for subsequent functional testing and optimization of the toehold switches.

In-vitro validation of the Toehold Switches with miRNA Targets

Following double transformation of toehold switches and their corresponding miR trigger under 2-hour IPTG induction, miR-10b toehold switch 2 demonstrated the most significant increase in GFP expression compared to the control (no toehold plasmid). This was indicative of its strong ability to detect miR-10b. In contrast, the other 10b toehold candidates showed GFP expression levels comparable to the control, suggesting they were ineffective in detecting miR-10b. These findings were further supported by statistical analysis where miR-10b toehold switch 2 shows a significant statistical difference

compared to the control as well as the other double transformations conducted (Supplementary Tables 1-3). This trend was also observed over 30 minute time intervals, further solidifying the efficacy of miR-10b toehold switch 2 (Supplementary Figures 1-4).
With regard to the miR-155 toehold switches, all three constructs exhibited significant increases in GFP expression following double transformation with the miR-155 plasmid. Among these, 155 Toehold 7 produced the highest GFP expression, which indicated that it was the most effective candidate for miR-155 detection. Once again, statistical analysis further confirmed these observations as all of the double transformations with the exception of the off-target case, showed a significant statistical difference compared to the control case (Supplementary Table 4-6).
Finally, off-target testing using the top candidate switches for miR-10b and miR-155 showed no activation in the presence of non-specific miRNAs, which displayed the specificity of the toehold candidate in each case.

Design and Functional Verification of Linker Sequences using DeNovo DNA
To ensure the efficacy of our tandem promoters in our reporter module, the performance of different linker sequences was evaluated. Constructs incorporating long GC-rich or GC/AT linkers demonstrated high levels of undesired bidirectional transcription and increased noise in predicted TSS profiles. In contrast, shorter AT-rich linkers produced cleaner, more focused transcription initiation with minimal antisense transcriptional noise. Notably, the 10 bp AT-rich linker sequence **ATTATATAAT**, along with its complementary reverse strand **TAATATATTA**, yielded the most favorable unidirectional transcription profile across constructs tested. These linkers consistently promoted strong, singular peaks of transcription initiation with low background activity, aligning with the intended design of our AND-gated tandem promoter system. This strategic selection process improved the transcriptional fidelity of the system, minimizing any potential crosstalk and optimizing GFP expression under AND-gated conditions. This DeNovo DNA analysis was critical for refining plasmid architecture and ensuring high specificity and performance required for downstream diagnostic applications.

Future Work
Given the cloning challenges experienced while attempting to construct the reporter module, future work on this project primarily involves validating the 'AND-gate' logic of the reporter module and conducting experiments in our one-pot, cell-free reaction.

AND-Gating Validation
Upon successful cloning, the reporter module will be tested for transcriptional activity under 3 conditions in our cell free system: no sigma factors, ECF41_491 only, ECF42_4454 only and both sigma factors. This will ensure that our reporter system is only activated in the presence of both sigma factors. Following successful confirmation of AND-gate logic, we will measure the fluorescence output as a function of sigma factor concentration. This will ensure that our diagnostic platform can accurately detect the overexpression of the breast cancer associated miRNAs.

Benchmarking and Fine-Tuning Expression Levels
Future experiments will also involve establishing quantitative benchmarks and threshold values for fluorescence signals to clearly distinguish between healthy and disease states. Expression levels of reporter signals will be fine-tuned through careful titration and optimization, thus enhancing the specificity and sensitivity of our diagnostic platform.

One-Pot Reaction using cancer serum and miRNA mimics
Finally after both aspects of our system have been validated, we will conduct experiments in our one-pot reaction combining the activity of the toehold switches as well as the reporter module in the presence of miR-10b and miR-155 mimics as well as breast cancer serum from cell lines: MCF-7, MDA-MB-231 and MCF-10A.

Testing with Human Blood Samples
To advance towards clinical applicability, we plan to test the diagnostic system with real human blood samples (both diseased and healthy states). This step will further validate the feasibility of our system in clinical settings and provide insights into its diagnostic performance in real biological environments.

Limitations
While significant progress was made in the design and validation of toehold switch components, challenges were encountered during the cloning process for the complete reporter module. Specifically, the original pET28 backbone vector used for constructing the tandem promoter-GFP system was found to be contaminated, which hindered successful assembly. Despite multiple attempts at cloning and verification, contamination issues and unexpected background expression prevented full construction and functional testing of the final reporter module. Future work will focus on sourcing a clean backbone vector and optimizing cloning conditions to complete the diagnostic system.

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AUTHOR CONTRIBUTIONS

AS: Methodology, Validation, Investigation, validated research methods, tandem promoters research and validation, and figure creation.

CP: Conceptualization, miRNA detection and quantification methodology, Toehold switch generation research, miRNA expression plasmid design, reached out to authors of similar projects

IM: Methodology, Investigation, Data Curation, tandem promoters research, figure creation, tandem promoters experimental validation and verification

SM: Plasmid design for toehold switches and reporter modules, Toehold Switch generation, toehold switch validation and plasmid construct verifications, outline and planning.

ETHICS STATEMENT

This study was conducted in accordance with ethical guidelines for synthetic biology and molecular diagnostics research. The research protocol was reviewed and approved by the IRB at Duke University, ensuring compliance with biosafety regulations and ethical standards for working with recombinant DNA, bacterial systems, and in vitro diagnostic assays.

No animal or human subjects were directly involved in this study beyond the use of de-identified biological samples, and all experiments adhered to biosafety level 1 (BSL-1) or BSL-2 standards, as required for handling bacterial cultures and synthetic genetic constructs.

COMPETING INTERESTS

None.

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METHODS

Toehold Switch Design

The de novo RNA toehold switches targeting miR-10b and -155 were designed using two distinct strategies. First, the miR-155 toehold switch sequence, as described by Liu et al., was repurposed by replacing the Kozak sequence with the Shine-Dalgarno sequence to accommodate the bacterial translation mechanism of the PURExpress In Vitro Protein Synthesis Kit [15]. Additionally, the placement of the start codon was altered due to the RBS change. Non-essential bases were changed to ensure the correct folding of the toehold switch using the DinaMelt software. Using this constructed miR-155 toehold switch, a de novo miR-10b switch was designed by substituting the trigger region. The remaining toeholds were designed using the UMaryland iGEM toehold generator software [16]. The software requires six inputs: the leader sequence, vestigial sequence, ribosome binding site (RBS), linker, target sequence, and reporter sequence to generate the toehold switch sequence. Five different vestigial regions, previously identified in literature, were utilized. Additionally, a leader sequence (GGG), suggested in several studies, was included for some designs to assess its compatibility with the toehold structure. By utilizing different combinations of the inputs 6 toehold switches for each miR target were designed.

Simulation and Validation of Toehold Switches

Upon designing the 14 toehold switches, simulation and validation of the toeholds were conducted using DinaMelt for secondary structure prediction to confirm the accuracy of the toehold configurations [17]. Stability assessments were also performed with this software to ensure each toehold switch maintained its intended structure under 25°C conditions. Further in-silico validation was accomplished using the UMaryland iGEM software and ViennaRNA software tools, specifically RNAcoFold and RNAup, to simulate binding interactions and efficiencies, as well as to predict the free energy changes associated with these interactions [15]. Key metrics obtained from these analyses included changes in the minimum free energy of the unbound and bound states of the toeholds, the delta G of binding, and the total free energy of the binding interactions. These results facilitated the selection of 3 toehold switches for each miR target that showed the most promise for further experimental validation. Additionally, in-silico tests for off-target effects were performed targeting miR-21, miR-155, and miR-10b using the same softwares discussed above, ensuring specificity of the designed toeholds to their respective trigger sequences.

Plasmid Construction and Verification

Design of de novo plasmids for:

The expression of Toehold Switches:
For the design of miR-10b and miR-155 toehold switches, the backbone from Addgene Plasmid #111908 (originally used for bacterial expression of an IL-8 toehold switch sensor) was utilized [19]. PCR was used to amplify the IL-8 backbone, and Gibson Assembly was performed to insert our custom toehold switch sequences into the linearized plasmid backbone, replacing the original IL-8 toehold region. For initial testing, GFP was used as the reporter gene. For the final construct the GFP reporter gene will be substituted with sigma factor ECF41_491 and ECF42_4454 for miR-155 and miR-10b toehold respectively. The sequence for the sigma factors were sourced from Plasmid #49702 and #49703 correspondingly.

The expression of miRNA:
For the design of the miRNA expressing plasmids, the backbone from Addgene Plasmid #62226 was utilized and the mature miRNA sequence was incorporated by replacing the gRNA sequence via PCR and Gibson assembly [20].

The expression of our tandem promoter reporter module:
For the construction of the diagnostic system's reporter module, the tandem promoter design integrated two previously characterized promoters, P41_up1141 and P42_up4062, which are selectively inducible by sigma factors ECF41_491 and ECF42_4454, respectively [21]. These promoters, modified in their -35 and -10 elements to respond to the distinct sigma factors, will be synthesized and then cloned using Gibson Assembly into a suitable vector backbone (Addgene Plasmid #171649) [22]. The inducible promoters will be connected to each other using linker sequences that were tested and verified using DeNovo DNA software. The entire tandem promoter setup was positioned upstream of the codon-optimized luciferase gene, chosen as the reporter due to its high sensitivity and ease of quantification through luminescence. It is to be noted that for initial testing GFP was used as the reporter.

Verification of the functionality of the de novo plasmid constructs and linkers using DeNovo DNA
To ensure accurate and efficient transcription of the tandem promoter constructs, multiple linker sequences were computationally assessed using the DeNovo DNA transcription prediction software. Specifically, a panel of GC-rich, AT-rich, and mixed GC/AT linkers in both short (approximately 10-15 bp) and extended (approximately 20-25 bp) forms were evaluated to analyze their impact on transcription initiation rates and directionality across transcription start sites (TSSs). For each design, the software generated predicted transcription initiation rate profiles across the forward and reverse strands. Linkers were selected based on achieving strong, localized forward transcription peaks with minimal reverse transcription, optimizing the constructs for downstream GFP expression under AND-gated conditions.

Cloning of Reporter Module
PCR was then performed using the 2X HiFi Mastermix protocol on Addgene plasmid 62226 (PTargetF), Addgene plasmid 111908 (IL-8 toehold switch sensor) and PET28a using pre-designed forward and reverse primers to create backbones for our candidate toehold switch and miRNA gene blocks to be inserted. The DNA was first amplified at 98℃ for 10 seconds. For 25-35 cycles, the samples underwent a second denaturation step at 98℃ for 10 seconds. The annealing temperatures were set to 63℃ for the PET28 and IL-8 backbone and 66℃ for PTargetF backbone and extension at 72℃ for 3 minutes.

Following PCR, 5ul of each PCR product was run on a gel to confirm correct amplification. After confirmation of correct amplification and accurate bands on the gel, DpnI digest was conducted for each PCR sample. This step was executed in DpnI buffer as opposed to Q5 buffer in order to enhance the efficiency of the digest. Finally, the PCR products for each sample underwent a PCR cleanup using the DNA Clean-and-Concentrator kit.

Gibson Assembly and Transformation
Our toehold plasmid and miRNA expression plasmid constructs were assembled using Gibson assembly. 50 ng of backbone and a corresponding 3x molar concentration of insert were added to a Gibson reaction tube containing buffers and made up to 10 ul with water.
5ul of Gibson mix was added to 50ul of DH5α E Coli cells and subsequently heat shocked after 15 minutes on ice. An SOC outgrowth step was performed for an hour before plating the cells and placing in the 37°C incubator.

Colony Picking and Culturing
Following transformation, plates containing spectinomycin (50 ug/mL) and kanamycin (30 ug/mL) were incubated at 37 ℃ for 16-18 hours. Plates that were created include 6 toeholds and one control along with 2 miRNA and one control. Individual colonies were identified and picked using sterile pipette tips. Each selected colony was inoculated into 5 mL of LB broth supplemented with either spectinomycin, kanamycin, or both. Cultures were grown overnight at 37 ℃ shaking consistently at 250 rpm to ensure optimal plasmid propagation. This step was essential to isolate monoclonal populations containing the correctly transformed constructs for subsequent plasmid extraction.

Plasmid Miniprep
Purified plasmid DNA was extracted from the overnight bacterial culture using the Qiagen Miniprep Kit according to the manufacturer's protocol. Briefly, all 5 mL of culture was pelleted at 13,000 rpm for 1 minute, and the supernatant subsequently discarded. The pellet was then resuspended in P1 buffer, lysed with P2 buffer, and neutralized using N3 buffer. The resulting lysate was centrifuged for 6 minutes to remove cellular debris, and the supernatant with no particulate was transferred to the provided silica spin column. After washing with PE buffer, DNA was eluted in 35 uL of nuclease-free water. Concentration and purity of the extracted plasmid DNA was quantified using a Nanodrop. Miniprepped plasmids were subsequently utilized for downstream applications including sequence verification and expression analysis.

Gel Electrophoresis and Extraction
Amplified pET28a backbone PCR products were separated via agarose gel electrophoresis following Addgene's standard protocol. 1% agarose gel was prepared and was run at 150V for 30 minutes. The bands corresponding to the *pET28a* backbone were excised and pooled to increase DNA concentration. The pooled DNA was then purified using the QIAquick Gel Extraction Kit (Qiagen) according to the manufacturer's instructions.

Double-Transformation
Dilutions of the plasmids were prepared with a concentration of 2 ng/µL. DH5α cells were thawed on ice for ~5 minutes. 1 µL of each plasmid was added into 20 µL bacteria. The cells were left on ice for 5 minutes then heat shocked for 30 seconds at 42°C. The cells were returned directly to ice for 2 minutes. 80 µL of SOC solution was added to the bacteria. This was followed by incubation for 60 minutes at 37 °C and 300 rpm. Dry agar plate, supplemented with Spectinomycin(25 µg/mL) and kanamycin (30 µg/mL) in the 37 °C incubator. The cells were then plated by pipetting 100ul of cells to each plate.
This double transformation process was observed to be inefficient so we opted for a sequential double transformation where competent target miRNA cells were first created and subsequently transformed with the corresponding toehold switch plasmids.

METHODS FOR FUTURE WORK
Tandem Promoters & Gating Mechanism
To validate the functionality of the tandem promoter system, we will perform a cell-free transcription-translation (TX-TL) assay using *E. coli*-derived lysates. Plasmids containing tandem promoters regulating firefly luciferase expression will be introduced into the reaction, along with varying

concentrations of sigma factors ECF41_491 and ECF42_4454.

Three experimental conditions will be tested: (1) Both sigma factors present at low, medium, and high concentrations, (2) Only one sigma factor added, and (3) No sigma factors (negative control). Luminescence will be measured using a plate reader at 30-minute intervals over a 3-hour period, and data will be analyzed to assess promoter activation and gating specificity.

To evaluate the AND-gating mechanism, luminescence intensity will be compared across conditions, with the expectation that only the co-presence of both sigma factors would lead to significant reporter gene expression. Statistical analysis will be performed using one-way ANOVA to confirm significant differences in luminescence output across experimental groups.

One-Pot Reaction

Our detection system will take place in a one pot reaction wherein all the above components, including the blood from patients, the two toehold switches, the reporter module plasmid, and the contents of the *PURExpress In Vitro Protein Synthesis Kit* are added [23]. We will then incubate to allow for the recognition and binding of miRNA biomarkers and the subsequent signal cascade.

Supplementary Tables and Figures

Statistical Analysis of miR-10b Toehold Switches Experimental Validation

ANOVA
(One Factor)

Sources	*SS*	*df*	*MS*	*F*	*P value*	*Eta-sq*	*RMSSE*	*Omega Sq*
Between Groups	1.8099E+17	3	6.033E+16	1190.91498	6.126E-11	0.99776583	19.9241476	0.99664968
Within Groups	4.0526E+14	8	5.0658E+13					
Total	1.8139E+17	11	1.649E+16					

Table 1: One Factor ANOVA conducted on miR-10b switch GFP fluorescent readings following double transformation with miRNA 10b after 2 hours under IPTG induction.

DUNNETT'S TEST alpha 0.05

group	*mean*	*size*	*ss*	*df*	*d-crit*
Control	127162902	3	6.8944E+12		
10b T2	390163564	3	3.6699E+14		
10b T5	71333498.9	3	4.9831E+12		
10b T7	140245148	3	2.6401E+13		
		12	4.0526E+14	8	2.88

D-TEST

group	*mean*	*std err*	*d-stat*	*lower*	*upper*	*p-value*	*mean-crit*	*Cohen d*
10b T2	-263000662	5811375.06	45.2561845	-279737422	-246263901	0	16736760.2	36.9515199
10b T5	55829403.2	5811375.06	9.60691792	39092643	72566163.3	0	16736760.2	7.84401564
10b T7	-13082246	5811375.06	2.2511446	-29819006	3654514.61	1	16736760.2	1.83805187

Table 2: Dunnett's test conducted on miR-10b switch GFP fluorescent readings following double transformation with miRNA 10b after 2 hours under IPTG induction.

Statistical Analysis of miR-10b Toehold Switches Off Target Validation

T TEST: Two Independent Samples (Equal Var) Alpha 0.05

	std err	*t-stat*	*df*	*p-value*	*t-crit*	*lower*	*upper*	*sig*	*effect r*
One Tail	7947335.22	39.4972206	4	1.2274E-06	2.13184679			yes	0.99872043
Two Tail	7947335.22	39.4972206	4	2.4549E-06	2.77644511	291832312	335962992	yes	0.99872043

Table 3: T- Test comparing miR-10b switch GFP fluorescent readings following double transformation with miRNA 10b and an off target miRNA after 2 hours under IPTG induction.

Statistical Analysis of miR-155 Toehold Switches Experimental Validation

ANOVA
(One Factor)

Sources	*SS*	*df*	*MS*	*F*	*P value*	*Eta-sq*	*RMSSE*	*Omega Sq*
Between Groups	1.876E+18	3	6.2534E+17	2434.3215	3.5267E-12	0.99890575	28.4857947	0.99835885
Within Groups	2.0551E+15	8	2.5688E+14					
Total	1.8781E+18	11	1.7073E+17					

Table 4: One Factor ANOVA conducted on miR-155 switch GFP fluorescent readings following double transformation with miRNA 155 after 2 hours under IPTG induction.

DUNNETT'S TEST alpha 0.05

group	*mean*	*size*	*ss*	*df*	*d-crit*
Control	66170078.1	3	3.4766E+12		
155 T4	161902283	3	2.4793E+13		
155 T5	529743397	3	2.3246E+14		
155 T7	1073616536	3	1.7943E+15		
		12	2.0551E+15	8	2.88

D-TEST

group	*mean*	*std err*	*d-stat*	*lower*	*upper*	*p-value*	*mean-crit*	*Cohen d*
155 T4	-95732205	13086450.8	7.31536816	-133421183	-58043227	0	37688978.2	5.97297309
155 T5	-463573319	13086450.8	35.423915	-501262298	-425884341	0	37688978.2	28.9235055
155 T7	-1.007E+09	13086450.8	76.9839336	-1.045E+09	-969757479	0	37688978.2	62.8571186

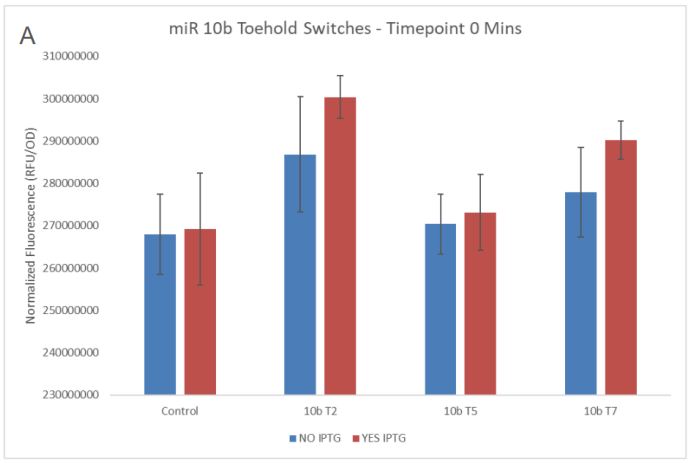
Table 5: Dunnett's test conducted on miR-155 switch GFP fluorescent readings following double transformation with miRNA 155 after 2 hours under IPTG induction.

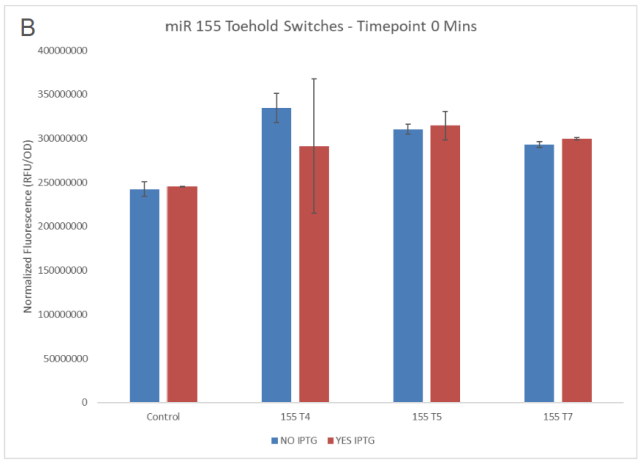
Statistical Analysis of miR-155 Toehold Switches Off Target Validation

T TEST: Two Independent Samples (Equal Var) Alpha 0.05

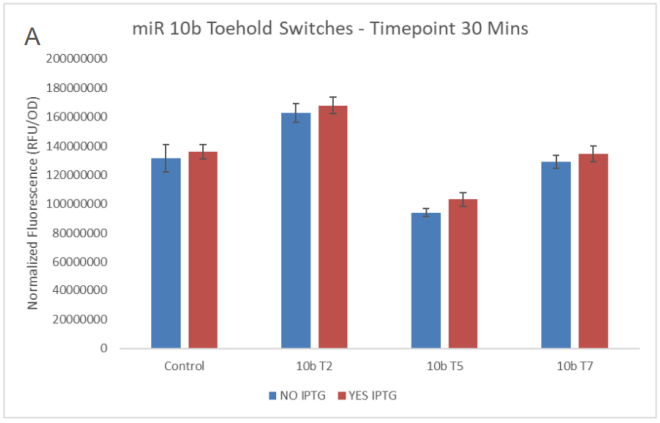
	std err	*t-stat*	*df*	*p-value*	*t-crit*	*lower*	*upper*	*sig*	*effect r*
One Tail	17403166.6	54.832729 5	4	3.3113E-07	2.13184679			yes	0.99933547
Two Tail	17403166.6	54.832729 5	4	6.6226E-07	2.77644511	905944192	100258206 5	yes	0.99933547

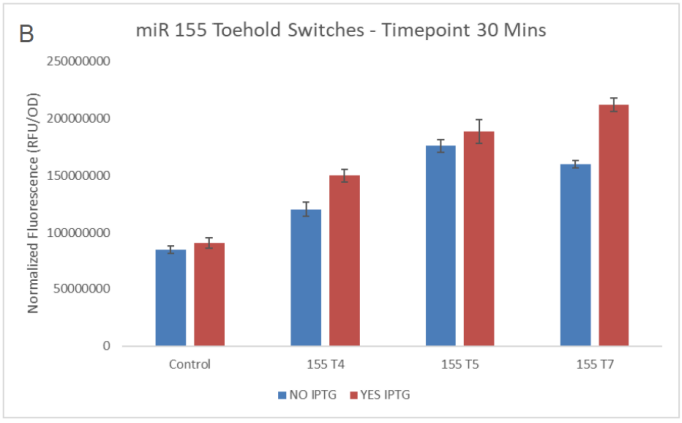
Table 6: T- Test comparing miR-155 switch GFP fluorescent readings following double transformation with miRNA 155 and an off target miRNA after 2 hours under IPTG induction.



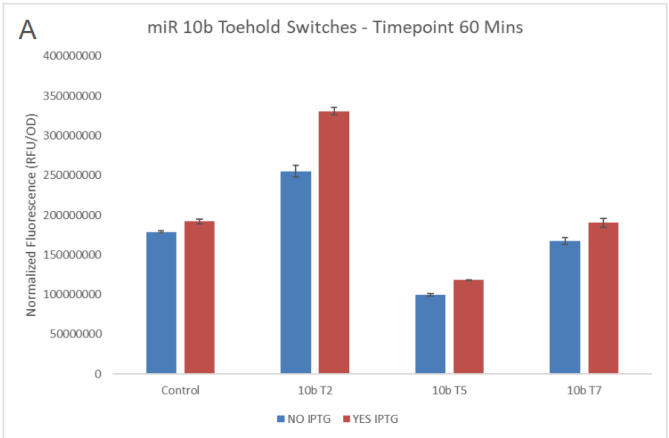


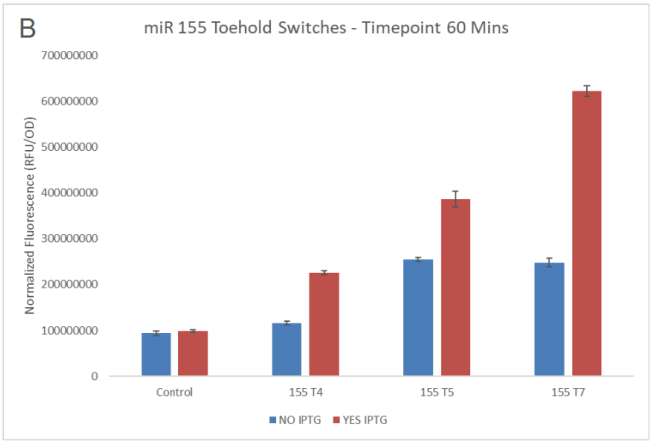
Supplementary Figure 1: GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 10b plasmids, normalized to OD600 at time point 0. **(B)** The level of GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 155b plasmids, normalized to OD600 at time point 0.





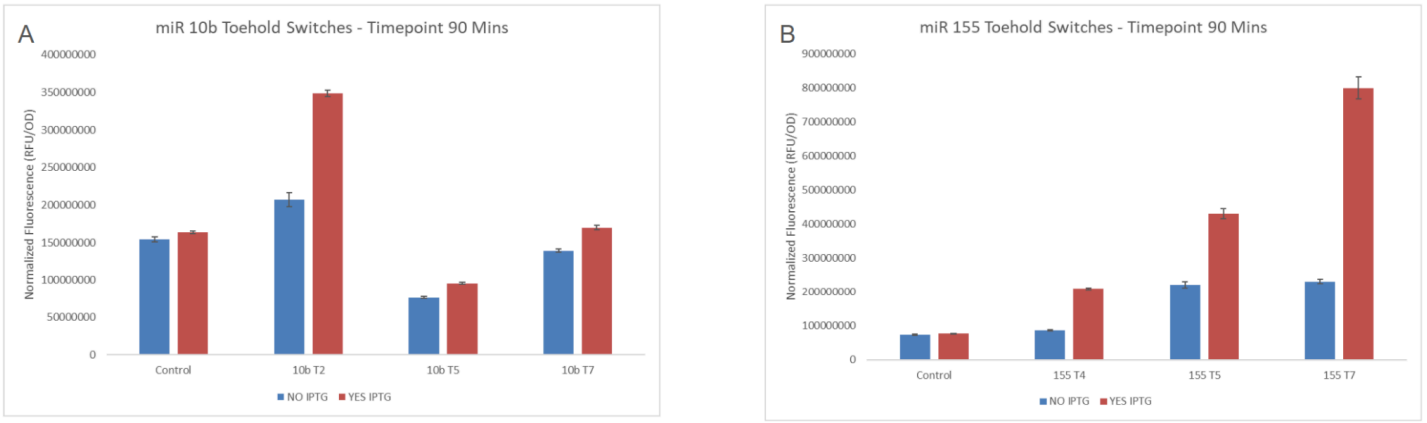
Supplementary Figure 2: GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 10b plasmids, normalized to OD600 after 30 minutes. **(B)** The level of GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 155b plasmids, normalized to OD600 after 30 minutes.



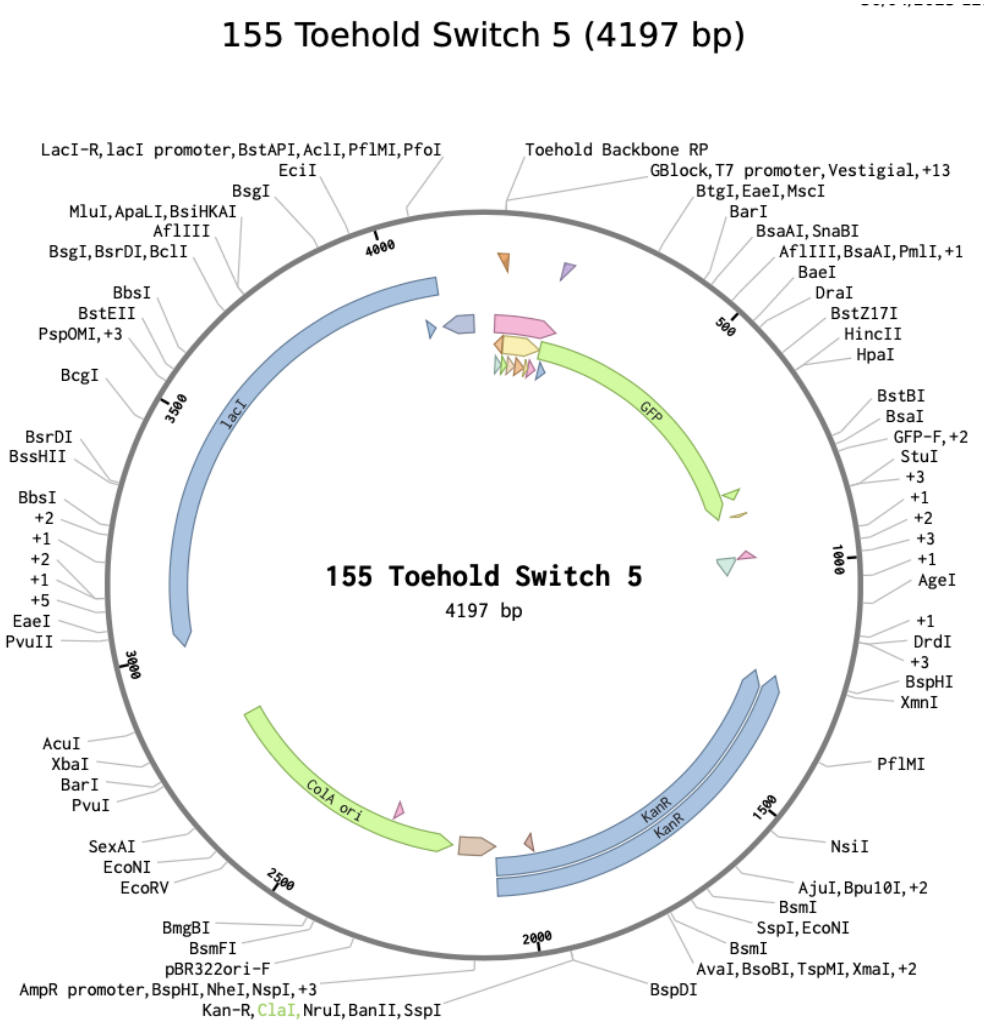


Supplementary Figure 3: GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 10b plasmids, normalized to OD600 after 60 minutes. **(B)** The level of GFP fluorescence

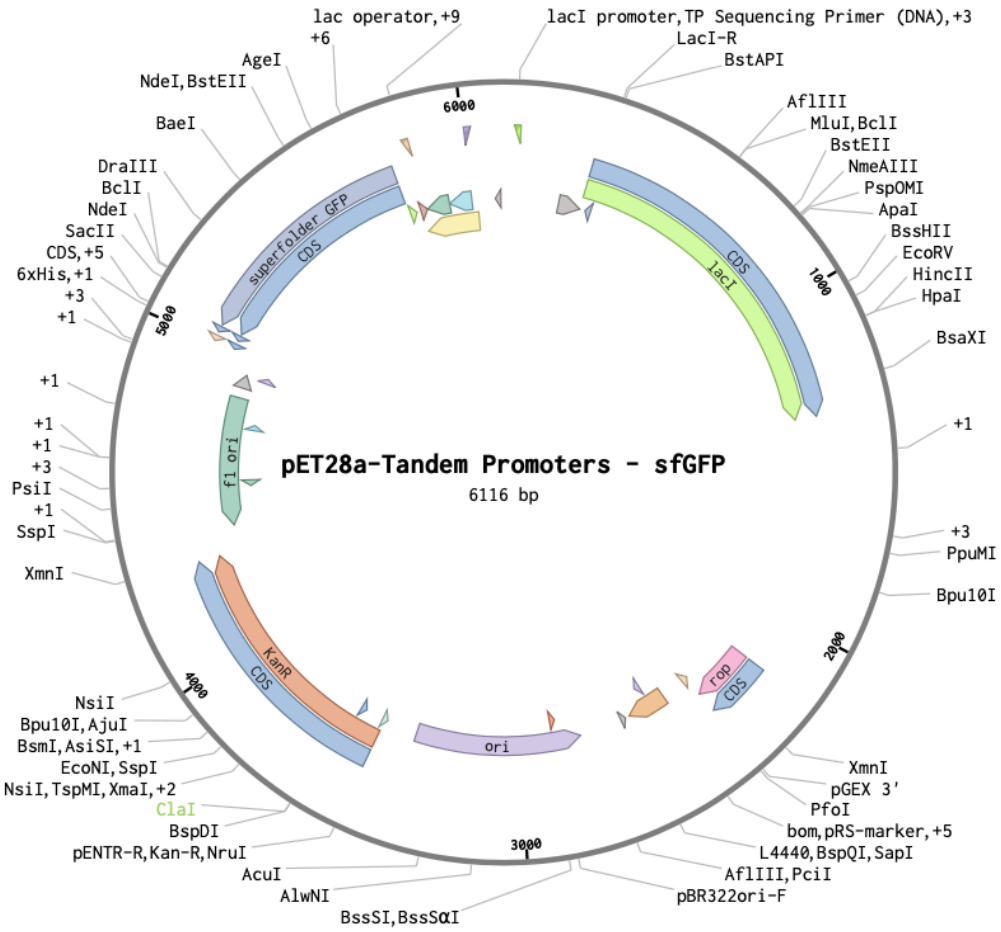
from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 155b plasmids, normalized to OD600 after 60 minutes.



Supplementary Figure 4: GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 10b plasmids, normalized to OD600 after 90 minutes. **(B)** The level of GFP fluorescence from bacterial colonies transformed sequentially with candidate toehold switch plasmids and microRNA 155b plasmids, normalized to OD600 after 90 minutes.



Supplementary Figure 5: Full plasmid map of toehold switches design developed on Benchling.
Link: https://benchling.com/s/seq-hEkmfMbPDUBFAGAhTIsY?m=slm-FP8VDMtirOyGf5ON7LP2



Supplementary Figure 6: Full plasmid map of tandem promoter reporter module design developed on Benchling.
Link: https://benchling.com/s/seq-ADBdWBjJJ8AmBHVBppcS?m=slm-LTFn71YPTzrns4IxHTrV